DSC5001: Statistical Machine Learning I

Tutorial 10: Bagging, Random Forests, and Gradient Boosting

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Task 1: Pharmacy Waiting Time

A pharmacy records the number of orders in its queue, x, and dispensing time, y (minutes).

Order ID	A	B	C	D	E	F
x	1	2	3	4	5	6
y	7	10	13	18	21	24

Two size-six bootstrap samples are

$$D^{*(1)} = (A, A, C, D, E, E), \qquad D^{*(2)} = (B, C, C, D, F, F).$$

Fit one regression stump to each sample, *using the fixed split* $x \leq 3.5$.

(a) Identify the out-of-bag (OOB) orders for each stump. Find the probability that a particular order is OOB in one size-six bootstrap sample.

(b) Calculate both stumps' leaf means and the bagged prediction at $x_* = 5$.

(c) Calculate OOB predictions and pooled OOB MSE over orders with at least one OOB prediction. Identify any orders with no OOB prediction.

(d) For a separate urgent-order classifier, three bagged trees give class-one probabilities $(0.85, 0.55, 0.05)$. Compare majority vote with probability averaging at threshold 0.5.

Solution 1(a): OOB Samples

A bootstrap sample draws six times *with replacement* from six original orders.

Tree	Sampled IDs	OOB IDs
1	A, A, C, D, E, E	$\boxed{B, F}$
2	B, C, C, D, F, F	$\boxed{A, E}$

One draw misses a specified order with probability $5/6$. All six draws miss it with probability

$$\boxed{q_6 = \left(\frac{5}{6}\right)^6 \approx 0.335.}$$

The actual OOB count in either displayed resample is two; q_6 is a *probability*, not a fixed fraction in every sample.

Solution 1(b): Bootstrap Stumps

Use $x \leq 3.5$ to form near and far leaves in *each resample*.

	Near $(x \leq 3.5)$		**Far** $(x > 3.5)$	
Tree	Responses	Mean	Responses	Mean
1	$(7, 7, 13)$	9	$(18, 21, 21)$	20
2	$(10, 13, 13)$	12	$(18, 24, 24)$	22

At $x_* = 5$, both stumps use their far leaf:

$$\widehat{f}^{*(1)}(5) = 20, \qquad \widehat{f}^{*(2)}(5) = 22.$$

The bagged regression prediction is

$$\boxed{\widehat{f}_{\text{bag}}(5) = \frac{20 + 22}{2} = 21 \text{ minutes.}}$$

The fixed threshold isolates the effect of *resampling* on predictions.

Solution 1(c): OOB Predictions

An order is evaluated only with a tree that *did not* train on it.

Order	**OOB tree**	y	$\widehat{y}_{\mathrm{OOB}}$	$(y-\widehat{y})^2$
B	1	10	9	1
F	1	24	20	16
A	2	7	12	25
E	2	21	22	1

The pooled MSE across these four orders is

$$\boxed{\mathrm{MSE}_{\mathrm{OOB}} = \frac{1+16+25+1}{4} = 10.75.}$$

Orders $\boxed{C, D}$ appear in *both* training resamples and have no OOB prediction here. Many more trees would improve OOB coverage.

Solution 1(d): Classification Aggregation

Each tree predicts class 1 when its class-one probability is at least 0.5.

$$(p_1, p_2, p_3) = (0.85, 0.55, 0.05) \quad \Longrightarrow \quad (\widehat{y}_1, \widehat{y}_2, \widehat{y}_3) = (1, 1, 0).$$

Majority vote: two positive votes, hence $\boxed{\widehat{y} = 1}$.

Probability average:

$$\overline{p} = \frac{0.85 + 0.55 + 0.05}{3} = \frac{1.45}{3} \approx 0.483.$$

At the same threshold, $\boxed{\widehat{y} = 0}$.

Voting counts class decisions; probability averaging retains their magnitudes. The two rules need not agree.

Task 2: Warehouse Packing Time

A warehouse predicts packing time y (minutes) from three binary features: heavy parcel H, gift wrapping G, and protective packaging P. Value 1 means the feature is present.

Order	1	2	3	4	5	6	7	8
H	0	0	0	0	1	1	1	1
G	0	0	1	1	0	0	1	1
P	0	1	0	1	0	1	0	1
y	10	12	14	16	18	20	22	24

(a) Compute root RSS and the child RSS for a split on each feature. Which feature would CART select?

(b) A forest samples $m_{\text{try}} = 2$ of the $q = 3$ features at each node. List the possible pairs and their best splits. Find the probability that H is absent from all four independently sampled root subsets.

(c) Derive the variance of the average of B equally variable, equally correlated trees. For $B = 5$, $v = 25$, compare correlation $\rho = 0.6$ (bagging) and $\rho = 0.2$ (forest).

(d) Validation RMSE is 3.0 before shuffling features and 6.2, 3.8, 3.2 after separately shuffling H, G, P. Calculate permutation importances and interpret them.

Solution 2(a): Packing-Time RSS

The root mean is $\overline{y} = (10 + 12 + \cdots + 24)/8 = 17$.

$$RSS_{\text{root}} = \sum_{i=1}^{8} (y_i - 17)^2 = \boxed{168}.$$

Heavy parcel H:

	$H = 0$	$H = 1$
Times	$(10, 12, 14, 16)$	$(18, 20, 22, 24)$
Mean	13	21
RSS	20	20

Hence $RSS_H = 20 + 20 = \boxed{40}$ and

$$\boxed{\Delta RSS_H = 168 - 40 = 128.}$$

Solution 2(a): Split Comparison

Compute within-child RSS in the same way for the other binary features.

Feature	Leaf means	Child RSS	Total RSS	Gain
H	$13, 21$	$20, 20$	40	$\boxed{128}$
G	$15, 19$	$68, 68$	136	32
P	$16, 18$	$80, 80$	160	8

All splits start with root RSS 168; larger gain is better.

CART chooses H when all three features are available.

The next question is why a forest sometimes *withholds* this strongest predictor.

Solution 2(b): Feature Sampling

With $q = 3$ features and $m_{\text{try}} = 2$, the three pairs are equally likely:

Available pair	Best RSS-reducing feature
$\{H, G\}$	H (gain 128)
$\{H, P\}$	H (gain 128)
$\{G, P\}$	G (gain 32)

Therefore $\Pr(H \text{ available}) = 2/3$ and $\Pr(H \text{ absent}) = 1/3$.
For four *independent* root-subset draws,

$$\boxed{\Pr(H \text{ absent in all four}) = \left(\frac{1}{3}\right)^4 = \frac{1}{81}.}$$

Feature randomness makes some trees consider alternatives; the split is still optimized within each subset.

Solution 2(c): Averaging Variance

For a fixed new input, let Z_b be the random prediction from tree b and define $\overline{Z}_B = B^{-1}\sum_{b=1}^{B} Z_b$. Assume $\mathrm{Var}(Z_b) = v$ and $\mathrm{Cov}(Z_b, Z_c) = \rho v$ for $b \neq c$.

$$\begin{aligned}
\mathrm{Var}(\overline{Z}_B) &= \frac{1}{B^2}\left[\sum_{b} \mathrm{Var}(Z_b) + \sum_{b\neq c} \mathrm{Cov}(Z_b, Z_c)\right] \\
&= \frac{Bv + B(B-1)\rho v}{B^2} \\
&= \boxed{v\left[\rho + \frac{1-\rho}{B}\right]}.
\end{aligned}$$

Correlation prevents the variance from declining as fast as v/B.

Solution 2(c): Numerical Comparison

Let $B = 5$, each tree's variance $v = 25$ min^2.
Bagging: $\rho = 0.6$.

$$\mathrm{Var}(\overline{Z}_{\mathrm{bag}}) = 25\left(0.6 + \frac{0.4}{5}\right) = \boxed{17 \text{ min}^2}.$$

Forest: $\rho = 0.2$.

$$\mathrm{Var}(\overline{Z}_{\mathrm{RF}}) = 25\left(0.2 + \frac{0.8}{5}\right) = \boxed{9 \text{ min}^2}.$$

Under the stated equal-variance assumptions, reduced correlation lowers the ensemble variance by $\boxed{8 \text{ min}^2}$.

Solution 2(d): Permutation Importance

For RMSE, *higher* error is worse. Define importance as

$$I_j = \mathrm{RMSE}_{\text{after shuffling } j} - \mathrm{RMSE}_{\text{original}}.$$

Feature	**RMSE after shuffle**	**Increase from** 3.0
H	6.2	$\boxed{3.2}$
G	3.8	$\boxed{0.8}$
P	3.2	$\boxed{0.2}$

The model relies most on H *for these validation cases*.
This measures predictive reliance, not causation; correlated features can also distort the interpretation.

Task 3: Greenhouse Water Demand

A greenhouse predicts daily water use y (liters) from sunlight hours x:

x	1	2	3	4	5	6
y	4	7	10	14	15	16

Use squared-error gradient boosting. Start from the sample mean F_0, fit two regression stumps to the current residuals, and use learning rate $\nu = 1/2$.
Candidate cutoffs are $1.5, 2.5, 3.5, 4.5, 5.5$.

(a) Find F_0, initial residuals and RSS. Calculate each candidate stump's within-leaf residual RSS and choose the first split.

(b) Fit the first stump h_1, update $F_1 = F_0 + \nu h_1$, and calculate its training RSS.

(c) Fit the best second stump h_2 to the *new* residuals. Calculate $F_2 = F_1 + \nu h_2$ and its training RSS.

(d) Validation pairs (x, y) are $(1.5, 6)$, $(3.3, 11)$, $(5.5, 17)$. Compare validation MSEs of F_1 and F_2 without refitting.

Solution 3(a): Initial Model

For squared error, the best constant prediction is the sample mean:

$$\boxed{F_0(x) = \frac{4 + 7 + 10 + 14 + 15 + 16}{6} = 11.}$$

The initial residual vector, in increasing-x order, is

$$\boxed{\boldsymbol{r}_0 = \boldsymbol{y} - 11 = (-7, -4, -1, 3, 4, 5).}$$

The initial residual sum of squares is

$$\boxed{RSS_0 = 49 + 16 + 1 + 9 + 16 + 25 = 116.}$$

A stump fits *one mean residual per leaf*. We compare all adjacent midpoints before choosing its split.

Solution 3(a): First Stump Search

For each cutoff s, minimize squared error separately in the groups $x \le s$ and $x > s$.

Cutoff s	1.5	2.5	3.5	4.5	5.5
Within-leaf RSS	$286/5$	$101/4$	$\mathbf{20}$	$221/4$	86

For $s = 3.5$, the residual groups are $(-7, -4, -1)$ and $(3, 4, 5)$:

$$\overline{r}_L = -4, \qquad \overline{r}_R = 4,$$

$$RSS_{\text{stump}} = [(-3)^2 + 0^2 + 3^2] + [(-1)^2 + 0^2 + 1^2] = \boxed{20}.$$

Choose the first split $\boxed{s_1 = 3.5}$.

Solution 3(b): First Boosting Round

The first fitted residual stump is

$$\boxed{h_1(x) = \begin{cases} -4, & x \le 3.5, \\ 4, & x > 3.5. \end{cases}}$$

With $\nu = 1/2$, predictions after Stage 1 are

$$\boldsymbol{F}^{(1)} = 11 + \frac{1}{2}(-4, -4, -4, 4, 4, 4) = \boxed{(9, 9, 9, 13, 13, 13)}.$$

The new residuals and training RSS are

$$\boldsymbol{r}_1 = \boldsymbol{y} - \boldsymbol{F}^{(1)} = \boxed{(-5, -2, 1, 1, 2, 3)},$$

$$\boxed{RSS_1 = 25 + 4 + 1 + 1 + 4 + 9 = 44.}$$

Solution 3(c): Second Stump Search

Fit the second stump to $\boldsymbol{r}_1 = (-5, -2, 1, 1, 2, 3)$, *not* to the original y.

Cutoff s	1.5	2.5	3.5	4.5	5.5
Within-leaf RSS	14	$\mathbf{29/4}$	20	$101/4$	$166/5$

The best second cutoff is $\boxed{s_2 = 2.5}$, *different* from the first cut 3.5.
The corresponding correction tree has leaf means

$$\overline{r}_L = \frac{-5-2}{2} = -\frac{7}{2}, \qquad \overline{r}_R = \frac{1+1+2+3}{4} = \frac{7}{4}.$$

Therefore

$$\boxed{h_2(x) = \begin{cases} -7/2, & x \le 2.5, \\ 7/4, & x > 2.5. \end{cases}}$$

Solution 3(c): Second Update

Add half the second stump to the *existing* model:

$$F_2(x) = F_1(x) + \frac{1}{2}h_2(x).$$

At the six training inputs, the predictions are

$$\boxed{\boldsymbol{F}^{(2)} = \left(\frac{29}{4}, \frac{29}{4}, \frac{79}{8}, \frac{111}{8}, \frac{111}{8}, \frac{111}{8}\right).}$$

Thus

$$\boldsymbol{r}_2 = \boldsymbol{y} - \boldsymbol{F}^{(2)} = \left(-\frac{13}{4}, -\frac{1}{4}, \frac{1}{8}, \frac{1}{8}, \frac{9}{8}, \frac{17}{8}\right).$$

$$\boxed{RSS_2 = \sum_{i=1}^{6} r_{i,2}^2 = \frac{263}{16} = 16.4375.}$$

Training RSS changes $116 \to 44 \to 16.4375$, but validation decides which model to keep.

Solution 3(d): Validation Error

Use the given validation inputs without refitting either model.

x	1.5	3.3	5.5
Actual y	6	11	17
$F_1(x)$	9	9	13
$F_2(x)$	$29/4$	$79/8$	$111/8$

The validation MSEs are

$$\mathrm{MSE}_{\mathrm{val}}(F_1) = \frac{(6-9)^2 + (11-9)^2 + (17-13)^2}{3} = \boxed{\frac{29}{3} \approx 9.667},$$

$$\mathrm{MSE}_{\mathrm{val}}(F_2) = \frac{(6-29/4)^2 + (11-79/8)^2 + (17-111/8)^2}{3} = \boxed{\frac{403}{96} \approx 4.198}.$$

Select F_2 between these two candidates. The lower training RSS alone would not justify the choice.

Task 4: Equipment Fault Alerts

An equipment monitor predicts fault status $y \in \{0, 1\}$ from sensor score x:

x	1	2	3	4	5
y	0	0	1	0	1

Let F be a *log-odds score*, $p = \sigma(F) = 1/(1 + e^{-F})$ the fault probability, and

$$\ell(y, F) = -y \log p - (1 - y) \log(1 - p).$$

Start from the observed class proportion, use the fixed stump split $x \leq 2.5$, and update with $\nu = 0.6$ and step multiplier $\gamma_1 = 1$.

(a) Derive $-\partial \ell / \partial F = y - p$. Find the initial constant score F_0 and summed training loss L_0.

(b) Compute initial pseudo-residuals, the two stump leaf values, and the updated scores F_1.

(c) Find the two updated leaf probabilities and total training loss L_1. Explain whether reduced loss requires every predicted class to change.

(d) Validation mean log-loss after rounds $M = 0, 1, 2, 3, 4$ is $(0.67, 0.61, 0.58, 0.60, 0.65)$. Choose the number of rounds and state the role of the final test set.

Solution 4(a): Logistic Gradient

The sigmoid derivative is $dp/dF = p(1-p)$. Applying the chain rule,

$$\begin{aligned}\frac{\partial \ell}{\partial F} &= \left(-\frac{y}{p} + \frac{1-y}{1-p}\right) p(1-p) \\ &= -y(1-p) + (1-y)p \\ &= p - y.\end{aligned}$$

Thus the negative gradient, or *pseudo-residual*, is

$$\boxed{r = -\frac{\partial \ell}{\partial F} = y - p.}$$

This differs from an ordinary numerical-response residual: F is a log-odds score, while p is the predicted probability.

Solution 4(a): Initial Model

Two of five fault labels equal 1, so the initial fault probability is

$$\boxed{p_0 = \frac{2}{5} = 0.4.}$$

The corresponding constant log-odds score is

$$\boxed{F_0 = \log \frac{p_0}{1 - p_0} = \log \frac{2}{3} \approx -0.4055.}$$

There are two positive labels and three negative labels. Their total cross-entropy is

$$\begin{aligned} L_0 &= -2 \log(0.4) - 3 \log(0.6) \\ &= \boxed{3.3651}. \end{aligned}$$

This L_0 is the *summed training loss*; the validation table in part (d) reports *mean* log-loss.

Solution 4(b): Logistic Stump

At $p_0 = 0.4$, the five pseudo-residuals are

$$\boxed{\boldsymbol{r}^{(0)} = \boldsymbol{y} - 0.4 = (-0.4, -0.4, 0.6, -0.4, 0.6).}$$

For the given split $x \leq 2.5$, the stump fits the *mean pseudo-residual* in each leaf:

$$h_L = \frac{-0.4 - 0.4}{2} = \boxed{-0.4},$$
$$h_R = \frac{0.6 - 0.4 + 0.6}{3} = \boxed{\frac{4}{15}}.$$

With $\nu = 0.6$, $\gamma_1 = 1$, update the *scores*:

$$\boxed{F_{1,L} = \log(2/3) - 0.24, \quad F_{1,R} = \log(2/3) + 0.16.}$$

Solution 4(c): Updated Probabilities

Convert the two updated log-odds scores to fault probabilities:

$$p_L = \sigma(\log(2/3) - 0.24) = \boxed{0.3440},$$
$$p_R = \sigma(\log(2/3) + 0.16) = \boxed{0.4389}.$$

The two left cases are both class 0. The right cases are $(1, 0, 1)$.
The updated *summed* logistic loss is

$$L_1 = -2\log(1 - p_L) - 2\log p_R - \log(1 - p_R)$$
$$= \boxed{3.0679} \; < \; L_0 = 3.3651.$$

At threshold 0.5, both leaves still predict class 0.
Lower cross-entropy need not change any hard classification decision.

Solution 4(d): Round Selection

Let M be the number of boosting rounds. The provided numbers are *mean validation log-loss* (lower is better).

M	0	1	2	3	4
Validation log-loss	0.67	0.61	**0.58**	0.60	0.65

Select

$$\boxed{M = 2 \text{ rounds.}}$$

Retain the model from this checkpoint; further rounds worsen validation loss in this example.
Evaluate the completed selected procedure on an *untouched final test set*. Do not use the test set to choose M.

Tutorial 10: ensemble construction, error correction, and evaluation

Questions and discussion